MA 324 | End-Semester Examination
Statistical Inference and Multivariate Analysis | 14:00–17:00 IST
IIT GUWAHATI | May 8, 2026

PART-B

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Signature of the Invigilator: Rayan

Instructions

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	0	7	5	1	13

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{y\boldsymbol{X}} \\ S_{y\boldsymbol{X}} & S_{\boldsymbol{XX}} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1\times 1}$ and $\boldsymbol{X}_{q\times 1}$ based on a centered (with respect to sample mean) sample of size n. Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector $\boldsymbol{X}$.

Soln:- first canonical correlation $\underline{\text{is}}$ $\text{corr}(X,Y) = \frac{\text{cov}(X,Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$

$$= \frac{S_{yX}}{\sqrt{S_{XX} S_{yy}}}$$

$$R^2 = \frac{SS_{Reg}}{SS_T}$$

$$SS_T = \sum_i (y_i - \bar{y})^2 = \cancel{S_{yy}} \cdot \text{Var}(Y) = S_{yy}$$

~~Consid~~ considering one of ~~multiple~~ linear regression of Y on vector X

~~then~~ then,

$$SS_{Reg} = \sum_i (\hat{y}_i - \bar{y})^2 = \sum_i (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2$$

$$= \sum_i (\hat{\beta}_0 \cancel{\bar{y}} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \cancel{\bar{y}})^2$$

$$= \sum_i (x_i - \bar{x})^2 \hat{\beta}_1^2$$

$$= S_{xx} \hat{\beta}_1^2$$

$$= \frac{S_{xy}^2}{S_{xx}} \quad \left(\because \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}\right)$$

$$SS_{reg} = \frac{S_{xy}^2}{S_{xx}} = \frac{S_{yx}^2}{S_{xx}}$$

Similarly for all one by one we get

$$SS_{reg} = \frac{S_{yx}^T S_{yx}}{S_{xx}}$$

$$\Rightarrow R^2 = \frac{S_{yx}^T S_{yx}}{S_{xx} S_{yy}}$$

Hence clearly,

$$r_1^2 = \frac{S_{yx}^T S_{yx}}{S_{xx} S_{yy}}$$

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So, $r_1^2 = R^2$.

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.

   (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.

   (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.

   (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^{p} \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

Soln:- (a) ~~Because first~~ for $m=2$, take two largest eigenvalues

ie $\lambda_3 = 4.2$, $\lambda_2 = 2.1$

$\lambda_3 \downarrow e_3$, $\lambda_2 \downarrow e_2$ → corresponding eigenvectors

$$\tilde{L} = \left[\sqrt{4.2}\, e_3 \;\middle|\; \sqrt{2.1}\, e_2 \right]$$

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(b) $\tilde{h}_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2$

$= (0.8)^2 + (-0.4)^2$

$= 0.64 + 0.16 = 0.8$

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(c) we know that,

$$\tilde{h}_i^2 = \tilde{l}_{i1}^2 + \tilde{l}_{i2}^2 \quad (\text{for } m=2)$$

$$\sum_{i=1}^{p} \tilde{h}_i^2 = \sum_{i=1}^{p} \tilde{l}_{i1}^2 + \sum_{i=1}^{p} \tilde{l}_{i2}^2$$

$$\tilde{L} = \left[\sqrt{\hat{\lambda}_1}\, \hat{e}_1 \;\middle|\; \sqrt{\hat{\lambda}_2}\, \hat{e}_2 \right]$$

first largest two eigenvalues of $\underline{S}$.

$\hat{e}_1, \hat{e}_2$ - their corresponding eigenvectors

$\tilde{L}$ has two columns,

clearly

$\sum_{i=1}^{p} \tilde{l}_{i1}^2$ → represent first column of $\tilde{L}\tilde{L}^T$

$$= \hat{\lambda}_1 \hat{e}_1 \hat{e}_1^T$$

$\because \hat{e}_1$ are normalized so, $\hat{e}_1 \hat{e}_1^T = 1$

$$= \hat{\lambda}_1.$$

Similarly. $\sum_{i=1}^{p} \tilde{l}_{i2}^2 = \hat{\lambda}_2$ → represent second column of $\tilde{L}\tilde{L}^T$

So, $\sum_{i=1}^{p} \tilde{h}_i^2 = \hat{\lambda}_1 + \hat{\lambda}_2 = \lambda_3 + \lambda_2 = 4.2 + 2.1$

$$= \underline{6.3}$$

first largest two.

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3. (5 points) Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Derive a variance-stabilizing transformation $g(\overline{X}_n)$ such that the asymptotic variance of $g(\overline{X}_n)$ does not depend on μ.

Soln:-

~~we know that~~

$\cancel{\sqrt{n}(\bar{X}_n - \mu) \rightarrow N(0,}$

$E(X_i) = \mu$

$E(\bar{X}_n) = \frac{1}{n} n\mu = \mu$

$Var(X_i) = \mu^2$

$Var(\bar{X}_n) = \frac{1}{n^2} n\mu^2$

$= \frac{\mu^2}{n}$

we know that,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N\left(0, \frac{\mu^2}{n}\right)$$

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N\left(0, [g'(\mu)]^2 \frac{\mu^2}{n}\right)$$

$$[g'(\mu)]^2 \frac{\mu^2}{n} = \text{constant (let } C)$$

$$g'(\mu) = \sqrt{\frac{Cn}{\mu^2}} = \frac{\sqrt{nC}}{\mu}$$

$$g(\mu) = \int \frac{\sqrt{nC}}{\mu} d\mu = \sqrt{nC} \ln \mu + \text{Constant}$$

$$g(\bar{X}_n) = \sqrt{nC} \ln \bar{X}_n + \text{constant}$$

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N\left(0, \left(\frac{\sqrt{nC}}{\mu}\right)^2 \frac{\mu^2}{n}\right)$$

$$= N\left(0, \frac{nC}{\mu^2} \frac{\mu^2}{n}\right)$$

$$= N(0, C)$$

Hence, variance does not depend on μ.

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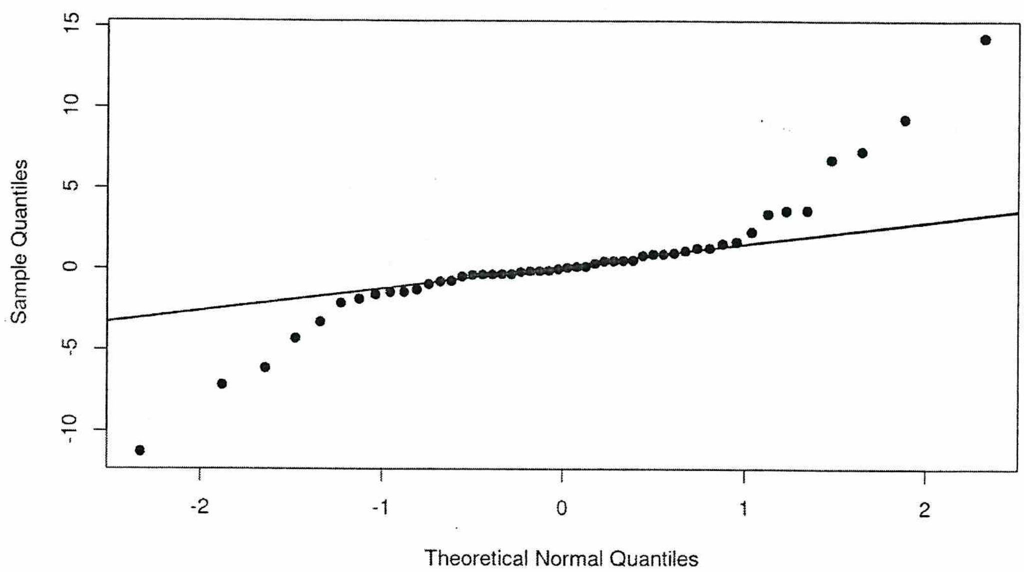
4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



Soln:-

~~Bec~~ If ql Points lie on ~~Straing~~ Straight line then it will follow normal distribution,

So, For approximatly from (-1, 1) it follows normal distribution.

what does it mean?)

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Additional Pages for Answers